The cubical lower bound theorem

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Abstract. We prove Jockusch's 1993 lower bound conjecture for cubical polytopes. A cubical d-polytope with n vertices and e edges satisfies $2e \geq (d+1)n - 2^d$ for every $d \geq 3$, and the corresponding lower bounds for all face numbers follow. Stacked cubical polytopes attain them. For $d \geq 4$, we strengthen the edge bound by $(d-1)$ times the imbalance between the two color classes of the graph. The proof shows that one facet determines every solution of a generic linear system on the graph: infinitesimal rigidity of the simplicial vertex figures propagates the solution across the boundary. The strengthened bound also holds for cubical PL spheres.

1. Introduction

A convex d-polytope is *cubical* if every facet is combinatorially a $(d-1)$-cube. In 1993, Jockusch [4] conjectured that the linear face-number bounds attained by stacked cubical polytopes hold for every cubical polytope. Klee [6] proved the edge bound for cubical 4-spheres in 2011, including the boundaries of cubical 5-polytopes. We prove the conjecture for every $d \geq 3$ and obtain an additional term measuring the graph's bipartition.

Write $f_i(P)$ for the number of i-faces. When $d \geq 3$, the graph of P is bipartite: its boundary sphere has zero first homology over $\mathbb{F}_2$, so every edge cycle is a sum of square boundaries and has even length. Let $B \sqcup W$ be its two color classes.

Theorem 1. *Every cubical convex d-polytope, $d \geq 3$, satisfies*

$$f_1(P) \geq \frac{d+1}{2} f_0(P) - 2^{d-1}.$$

For $d \geq 4$, the stronger inequality

$$f_1(P) \geq \frac{d+1}{2} f_0(P) - 2^{d-1} + \frac{d-1}{2}\big||B| - |W|\big|$$

holds. The first bound is attained by cubes and stacked cubical polytopes.

The edge inequality implies all the conjectured face-number bounds by the cubical McMullen–Perles–Walkup reduction; these are stated in Corollary 5. The proof of the edge inequality uses the normalized $(1, d)$ system of bipartite rigidity [5], with parameters independent of the polytope's coordinates. Its solution space injects into the solution space on a single facet, whose dimension is 2^{d-1}. This gives the bound by counting unknowns and equations.

The injection is a local-to-global consequence of rigidity. At each vertex, infinitesimal rigidity of the simplicial vertex figure forces vanishing on one incident facet to propagate to every neighbor. Connectedness of the facet adjacency graph then propagates vanishing through the whole boundary. Convexity enters only through vertex-figure rigidity; generic rigidity of simplicial PL spheres gives the same strengthened bound for cubical PL spheres.

2. **Local propagation**

Assume $d \geq 4$. Choose vectors $p_w \in \mathbb{R}^d$, $w \in W$, and consider the linear system

$$p_w \cdot u_b + s_w = 0 \qquad (bw \in E(P)), \tag{1}$$

with unknowns $u_b \in \mathbb{R}^d$ and $s_w \in \mathbb{R}$. Denote its solution space by $M(P, p)$.

The unequal numbers of unknowns at the two colors are deliberate. If one facet determines a solution, the number of equations must be at least $d|B| + |W|$ minus the dimension of the solution space on that facet. To prove such determination, we must transfer information between black vertices. A square does exactly this: subtracting its two equations at the white vertex opposite w eliminates the scalar unknown and gives an equation for the difference of two black vectors. With a suitable choice of the parameters, these are the rigidity equations of the vertex figure at w.

We use the Dehn–Whiteley rigidity theorem: the vertices and edges of every convex simplicial k-polytope, $k \geq 3$, form an infinitesimally rigid framework in $\mathbb{R}^k$ [7, Theorems 3.1 and 8.6]. Thus any motion of its vertices has the form $v_a = t + Aq_a$, with $A^T = -A$. If the motion vanishes at the k vertices of one facet, then A annihilates their $(k-1)$-dimensional space of differences. Its rank is at most one and, being even, is zero. The whole motion consequently vanishes.

Lemma 2. *For generic p, the following holds for every white vertex w and every facet F containing it. If a solution of* (1) *has $u_b = 0$ at the neighbors of w in F, then $u_b = 0$ at every neighbor of w.*

Proof. Fix w and F. The vertex figure L at w is a simplicial $(d-1)$-polytope. Its vertices are indexed by the black neighbors of w, and the $d-1$ neighbors in F form a facet T of L. An edge ab of L corresponds to a square of P with cyclic order

$$w, a, z_{ab}, b, \qquad z_{ab} \in W.$$

Different edges have different opposite vertices z_{ab}. Otherwise two squares would intersect in a face containing their common opposite vertices w, z_{ab}, although no proper face of a square contains such a pair. The squares, and hence the edges of L, would coincide.

Pin $u_b = 0$ on T. An edge equation at w gives $s_w = 0$. The equations at w, and the differences of the two equations at each opposite square vertex z_{ab}, imply

$$p_w \cdot u_b = 0 \quad (b \in V(L)), \qquad p_{z_{ab}} \cdot (u_a - u_b) = 0 \quad (ab \in E(L)).$$

To show that this pinned system generically has zero kernel, it suffices to exhibit one specialization of full column rank. Choose the vertex coordinates $q_b \in \mathbb{R}^{d-1}$ of L and specialize

$$p_w = e_d, \qquad p_{z_{ab}} = (q_a - q_b, 0).$$

These assignments are consistent because the labels z_{ab} are distinct and different from w. The first equations put all u_b in $\mathbb{R}^{d-1} \times \{0\}$; the second become the infinitesimal rigidity equations of L. The pinned-facet consequence of the rigidity theorem forces every u_b to vanish.

A full-column minor of this local matrix is therefore a nonzero polynomial in the coordinates of all the p_w. There are finitely many pairs (w, F), so the product of one such minor for each pair is nonzero. Its nonvanishing gives all the assertions simultaneously. The specializations used to prove that the separate minors are nonzero need not agree: they establish nonempty open conditions, whose finite intersection is nonempty. □

We also choose p so that the $d-1$ vectors indexed by the white neighbors of any black vertex within any facet are independent. These are further nonempty polynomial open conditions, compatible with the lemma.

3. **Determination by one facet**

For a facet F, let $M(F, p)$ be the solution space of the equations restricted to its vertices and edges.

Proposition 3. *For generic p and every facet F, restriction gives an injection*

$$M(P, p) \longrightarrow M(F, p), \qquad \dim M(F, p) = 2^{d-1}.$$

Proof. Suppose a solution vanishes on F. Let F' be an adjacent facet and $R = F \cap F'$ their common ridge. At every white vertex of R, all black neighbors in F have zero vectors. Lemma 2 makes all its black neighbors in P zero. Within the cube F', every vertex outside the facet R is adjacent to its matched vertex in R. Thus all black vectors on F' vanish, and its edge equations then make all white scalars vanish. The facet adjacency graph is connected, so this propagates through the boundary and proves injectivity.

Put $s = 2^{d-2}$. The facet has s vertices of each color and $(d-1)s$ edges, giving $(d+1)s$ scalar unknowns. Its edge equations are independent. Indeed, a linear relation among their rows, restricted to the vector columns of a black vertex b, reads

$$\sum_{w:\, bw \in E(F)} \lambda_{bw} p_w = 0.$$

The $d-1$ vectors are independent by our choice of p, so every coefficient at b is zero. Repeating this at all black vertices proves row independence. Hence

$$\dim M(F,p) = (d+1)s - (d-1)s = 2^{d-1}. \qquad \square$$

Proof of Theorem 1. Write $e = f_1(P)$. System (1) has $d|B| + |W|$ scalar unknowns and e equations. Proposition 3 gives

$$d|B| + |W| - e \le \dim M(P,p) \le 2^{d-1}.$$

Interchanging B and W gives the second asymmetric bound. Their maximum is

$$\max\{d|B| + |W|,\ |B| + d|W|\} = \frac{d+1}{2} f_0(P) + \frac{d-1}{2} \big||B| - |W|\big|.$$

This proves the stated inequality for $d \ge 4$. For $d = 3$, quadrilateral incidence and Euler's formula give $2f_1 = 4f_2$ and $f_0 - f_1 + f_2 = 2$, hence $f_1 = 2f_0 - 4$. $\square$

Corollary 4. *The strengthened edge bound in Theorem 1 holds for every finite cubical PL $(d-1)$-sphere, $d \ge 4$, whose cells intersect in common faces.*

Proof. The link of each vertex is a simplicial $(d-2)$-sphere. By Fogelsanger's rigidity theorem [3, Theorem 3.14], its graph has an infinitesimally rigid generic placement in $\mathbb{R}^{d-1}$. Use this placement for the coordinates q_b in Lemma 2; its facet vertices are affinely independent, so the same pinning argument applies. The common-face condition again makes the opposite square vertices distinct. The graph is bipartite by the homology argument above, and the facet adjacency graph is connected. The rest of the proof is unchanged. $\square$

4. The remaining face numbers

The remaining bounds follow by the cubical McMullen–Perles–Walkup reduction observed by Jockusch and recalled in [1].

Corollary 5. *Let P be a cubical d-polytope, $d \geq 3$, with n vertices. Then*

$$f_i(P) \geq \frac{\binom{d}{i} + \binom{d-1}{i-1}}{2^i}\, n - 2^{d-i}\binom{d-1}{i-1} \qquad (1 \leq i \leq d-2),$$
$$f_{d-1}(P) \geq \frac{d-1}{2^{d-2}}\, n - 2(d-2).$$

All these bounds are attained by stacked cubical polytopes.

Proof. For $d \geq 4$, apply the simplicial lower bound theorem [2] to the $(d-1)$-dimensional vertex figures L_v and sum over vertices. Since an i-face has 2^i vertices, this gives, for $1 \leq i \leq d-2$,

$$2^i f_i(P) = \sum_v f_{i-1}(L_v) \geq 2\binom{d-1}{i-1} f_1(P) - (i-1)\binom{d}{i} n.$$

Substitute Theorem 1 and use Pascal's identity to obtain the first bound. For facets the corresponding simplicial bound is

$$f_{d-2}(L_v) \geq (d-2)\deg(v) - d(d-3).$$

Its sum gives $2^{d-1} f_{d-1}(P) \geq (2d-2)n - (d-2)2^d$. The dimension-three case follows from $f_1 = 2n-4$ and $f_2 = n-2$.

A d-cube attains every bound. Attaching another cube along a facet adds 2^{d-1} vertices, adds $2^{d-1-i}\big(\binom{d}{i} + \binom{d-1}{i-1}\big)$ i-faces for $1 \leq i \leq d-2$, and replaces one facet by $2d-1$ others. These are exactly the increments in the displayed lower bounds. □

For $d \geq 4$, the same substitution adds $2^{-i}(d-1)\binom{d-1}{i-1}\Delta$ to the i-face bound and $2^{1-d}(d-1)(d-2)\Delta$ to the facet bound, where $\Delta = \big||B| - |W|\big|$.

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